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ICT-106

I. Translate the given sentences into logical expressions.

Let p , q , and r be the propositions

p : Grizzly bears have been seen in the area.

q : Hiking is safe on the trail.

r : Berries are ripe along the trail.

a. Berries are ripe along the trail, but grizzly bears have not been seen in the area.

$$r \wedge \neg p$$

b. If berries are ripe along the trail, hiking is safe if and only if grizzly bears have not been seen in the area.

$$r \rightarrow (q \leftrightarrow \neg p)$$

c. Hiking is not safe on the trail whenever grizzly bears have been seen in the area and berries are ripe along the trail.

$$(p \wedge r) \rightarrow \neg q$$

2. Let p , q , and r be the propositions

p : You get an A on the final exam.

q : You do every activity in this class.

r : You get an A in this class.

a. You will get an A in this class unless you do not get an A on the final exam.

$$r \vee \neg p$$

b. You will get an A in this class if and only if you do every activity or you get an A on the final exam.

$$r \leftrightarrow (q \vee p)$$

II. Construct the truth tables for each compound proposition.

$$p \vee q \rightarrow \neg p \wedge \neg q$$

$$(p \rightarrow q) \oplus (\neg p \leftrightarrow q) \wedge \neg q$$

$$(p \leftrightarrow \neg q) \rightarrow \neg (q \vee r)$$

p	q	r	$p \vee q$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$	$p \vee q \rightarrow \neg p \wedge \neg q$	$p \rightarrow q$	$\neg p \leftrightarrow q$	$(p \rightarrow q) \oplus (\neg p \leftrightarrow q)$	$(p \rightarrow q) \oplus (\neg p \leftrightarrow q) \wedge \neg q$	$p \leftrightarrow \neg q$	$q \vee r$	$\neg(q \vee r)$	$(p \leftrightarrow \neg q) \rightarrow \neg(q \vee r)$
T	T	T	T	F	F	F	F	T	F	T	F	F	T	F	T
T	T	F	T	F	F	F	F	T	F	T	F	F	T	F	T
T	F	T	T	F	T	F	F	F	T	T	T	T	T	F	F
T	F	F	T	F	T	F	F	F	T	T	T	T	F	T	T
F	T	T	T	T	F	F	F	T	T	F	F	T	T	F	F
F	T	F	T	T	F	F	F	T	T	F	F	T	T	F	F
F	F	T	F	T	T	T	T	T	F	T	T	F	T	F	T
F	F	F	F	T	T	T	T	T	F	T	T	F	F	T	T